ChatGPT

Production-ready EMS study app database blueprint

Executive summary

Assumption: the learner's protocol locale is unspecified, so the database should default to the **US national model** and always carry a **local override layer** for state, agency, and medical-director differences. The safest backbone is: National EMS scope and education documents for level boundaries and curriculum; current public National Registry of Emergency Medical Technicians exam plans and handbooks for exam alignment; current American Heart Association guidance for resuscitation; National Association of State EMS Officials and EMS.gov evidence-based/model guideline material for protocol logic; and paediatric-readiness resources from Emergency Medical Services for Children for paediatric systems content. The Education Standards are minimum competency standards, not a one-size-fits-all curriculum, and NHTSA's scope model is a national framework rather than a substitute for local law or credentialling. [1]

The workbook therefore should be a **normalised relational spreadsheet**, not a single tab. It needs separate layers for taxonomy, lessons, foundations/revisit logic, scenarios, questions, media, references, protocol overrides, QC, and changelog. It must use **only original NREMT-style items**: NREMT explicitly states that exam materials are copyrighted and that disclosing, discussing, posting, or reconstructing exam items from memory is prohibited. Public NREMT materials also show that all levels now use technology-enhanced items, with AEMT/Paramedic heavily emphasising clinical judgement and integrated paediatric content. AHA's current guideline publishing model is modular, which fits workbook chunking well. [2]

For media, prioritise **public, primary datasets** over screenshots from random teaching sites: PhysioNet [3] for raw ECG and heart-sound data, and VitalDB [4] for de-identified multi-parameter monitor waveforms. Use paediatric-readiness resources from Health Resources and Services Administration [5] / EMSC primarily for policy, equipment, checklist, and readiness-training content rather than as the main source of waveform truth. Some asset sources have clear open licences, but anything with unclear redistribution rights should be marked `permission_required` or `link_only`. [6]

Master prompt

This prompt is tuned to the public exam blueprints from National Registry of Emergency Medical Technicians [7], the national scope and education framework published via National Highway Traffic Safety Administration [8] EMS resources, the current guidance structure from the American Heart Association [9], and the model-guideline/evidence-based work of National Association of State EMS Officials [10]. It is built to protect copyright, preserve stable IDs, and force a national-core plus local-override architecture. [11]

```
You are an EMS curriculum architect, assessment designer, media librarian,
and data engineer.
```

OBJECTIVE
Build a production-ready, multi-sheet Excel/Google Sheets workbook for an interactive EMS study app covering EMT, AEMT, and Paramedic education and exam preparation in the United States.

The workbook must support:
- lessons/tutorials
- foundations and revisit pathways
- flashcards
- interactive branching scenarios
- skills checklists
- drug cards
- ECG drills
- audio drills (heart/lung sounds where rights permit)
- medication-maths drills
- original NREMT-style question banks
- remediation plans
- mastery tracking
- media asset management
- export to XLSX, per-sheet CSV, and JSONL

ASSUMPTIONS
- User locale for protocol rules is unspecified.
- Use the US national model as the default spine.
- Add a local override layer for state, territory, agency, and medical-director differences.
- Ingest the user-provided Google Drive slide folder when accessible:
  https://drive.google.com/drive/folders/1f0IIZQRaXO2iXKqpxiBORYOzl2jT4V2I?usp=sharing

HARD RULES
- Do NOT fetch, copy, quote, reconstruct, or paraphrase copyrighted or memorised NREMT exam items.
- Create only original NREMT-style items aligned to current public NREMT test plans, handbooks, and sample-item concepts.
- Preserve stable IDs forever. Never reuse or rewrite previously issued IDs.
- If output is too large, continue in sequential chunks and append only.
- Keep the same schema across chunks.
- Use snake_case column names.
- No merged cells, no decorative formatting, no colour-dependent meaning.
- One row per atomic item whenever possible.
- All clinically meaningful rows must link to references.
- Flag unsupported, outdated, or protocol-dependent content instead of guessing.

SOURCE PRIORITY
1. National EMS Scope of Practice Model
2. National EMS Education Standards
3. EMT/AEMT/Paramedic Instructional Guidelines or companion official educational guidance
4. Current NREMT public test plans, handbooks, and sample packets

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5. Current AHA CPR/ECC and first aid guidance
6. NASEMSO National Model EMS Clinical Guidelines and EMS.gov evidence-based
guidance
7. HRSA/EMSC pediatric readiness resources
8. User-provided textbook slides and study materials
9. Other authoritative primary sources only when clearly needed

CONFLICT POLICY
- Scope-of-practice conflict: source #1 wins.
- Curriculum/coverage conflict: #2 and #3 win.
- Exam-alignment conflict: #4 wins.
- Resuscitation conflict: #5 wins.
- Protocol-process conflict: #6 informs but mark local_override_needed where
applicable.
- If same-tier sources conflict, prefer the newer document and set
source_conflict_flag = true.
- If unresolved, set verification_status = pending_manual_review.

REQUIRED SHEETS
At minimum produce:
readme
data_dictionary
taxonomy
modules
lessons
lesson_objectives
foundations
foundations_review_map
remediation_rules
skills
skill_steps
drugs
drug_classes
ecg_bank
calculation_drills
scenarios
scenario_steps
scenario_branches
questions
question_options
question_rationales
psychometrics
references
content_links
content_reference_links
media_manifest
media_links
protocol_overrides
glossary
slide_inventory
slide_mapping
```

```
qc_log
changelog
chunk_manifest

FOUNDATIONS / REVISIT SPIRAL
Create explicit foundations content for:
- anatomy_and_physiology
- medical_terminology
- abbreviations_and_acronyms
- pathophysiology
- pharmacology_basics
- medication_math
- communication_and_documentation
- legal_ethics_consent
- evidence_based_ems_basics
- safety_and_wellness
- life_span_development
- public_health

For every advanced lesson, scenario, skill, drug, ECG, drill, item, and
media-linked activity:
- link prerequisite foundation IDs
- define revisit_trigger logic
- define weak_area_followup actions
- explain why_this_foundation_matters_here

QUESTION RULES
- Original NREMT-style items only.
- Supported item types:
  single_best_answer
  multiple_response
  build_list
  drag_drop
  hotspot
  graphical_ecg
  waveform_interpretation
  scenario_cluster
  clinical_judgment
- Every question must include:
  question_id
  level
  domain
  subdomain
  topic
  item_type
  difficulty_1_to_5
  stem
  options
  correct_answer
  why_correct
  why_each_wrong
```

```
  reference linkage
  remediation linkage
  national_vs_local_flag
  psychometric/QC fields

SCENARIO RULES
- Create a large library of realistic, phase-based scenarios.
- Cover dispatch, en route, arrival, scene size-up, primary assessment,
immediate interventions, secondary assessment, differential refinement,
reassessment, transport, handoff, and debrief.
- Include age groups, settings, special populations, MCI/operations overlays,
documentation, handoff, and refusal cases.
- Over-represent paediatrics relative to raw call frequency because high-
quality practice requires deliberate repetition.

MEDIA RULES
- Prioritise public/primary sources first (PhysioNet, PTB-XL, MIT-BIH, CirCor
DigiScope, VitalDB, official pediatric readiness resources).
- Keep raw masters separate from app-ready derivatives.
- Do not redistribute any asset unless licence terms permit it.
- If licence is unclear, mark permission_required or link_only.
- Every media row must include attribution, licence fields, anonymisation
check, checksum, and linked content IDs.

OUTPUT RULES
Produce:
1. Multi-sheet workbook (XLSX and/or Google Sheets compatible)
2. Data dictionary tab
3. CSV export for each sheet
4. JSONL export for app import
5. Media manifest CSV and JSON
6. QC summary
7. PDF containing the polished master prompt and implementation report for
the Replit Agent

CHUNKING RULES
If content is too large:
- continue in sequential chunks
- preserve IDs and schema
- append only
- record every chunk in chunk_manifest
- never silently rewrite old rows
- issue explicit changelog entries for any correction
```

Workbook architecture

A spreadsheet that is meant to power an app should follow the same logic as the national EMS ecosystem: scope and education define the spine, exam blueprints define assessment emphasis, and local protocol sits on top as a variation layer. The 2021 National EMS Education Standards are minimum competencies rather than a day-by-day curriculum, which is exactly why a normalised workbook with

references, joins, and override flags is the safer design. AHA's 2025 guideline chapters are now organised into modular "knowledge chunks", which also supports chunked, app-ready content objects rather than giant textbook tabs. [12]

Workbook sheet list

Sheet	Purpose	Key columns
readme	import/use notes	workbook_version, schema_version, build_date
data_dictionary	schema registry	sheet_name, column_name, data_type, description
taxonomy	master hierarchy	taxonomy_id, parent_taxonomy_id, taxonomy_type, display_name
modules	top-level course groupings	module_id, taxonomy_id, level, title
lessons	atomic teaching units	lesson_id, module_id, taxonomy_id, title
lesson_objectives	one objective per row	objective_id, lesson_id, bloom_level
foundations	A&P/terminology/ pathophys/med maths bank	foundation_id, cluster, title
foundations_review_map	revisit/prereq map	frm_id, source_content_type, source_content_id, foundation_id
remediation_rules	auto-remediation logic	rule_id, trigger_type, target_content_id
skills	skill checklists	skill_id, title, level_min
skill_steps	ordered skill actions	skill_step_id, skill_id, step_order
drugs	medication cards	drug_id, generic_name, class_name
drug_classes	normalised drug class table	drug_class_id, class_name
ecg_bank	rhythm/12-lead items	ecg_id, rhythm_name, ecg_type
calculation_drills	maths drills	calc_id, calc_type, level_min
scenarios	scenario headers	scenario_id, taxonomy_id, level, setting
scenario_steps	phase-by-phase flow	step_id, scenario_id, phase
scenario_branches	decision branches	branch_id, from_step_id, to_step_id
questions	item stems	question_id, item_type, difficulty_1_to_5
question_options	option rows	option_id, question_id, option_key
question_rationales	right/wrong rationales	rationale_id, question_id, rationale_type
psychometrics	item QA/pilot metadata	psy_id, question_id, item_status
references	source library	reference_id, organisation, title

Sheet	Purpose	Key columns
content_links	generic many-to-many join	link_id, from_type, from_id, to_type, to_id
content_reference_links	citations by content row	cref_id, content_type, content_id, reference_id
media_manifest	import-ready asset catalogue	media_id, asset_type, source_name, file_name
media_links	media-to-content join	mlink_id, media_id, content_type, content_id
protocol_overrides	state/agency/local deltas	override_id, jurisdiction_name, content_id
glossary	terms and abbreviations	glossary_id, term, definition
slide_inventory	Google Drive slide ingest log	slide_file_id, file_name, ingest_status
slide_mapping	slide-to-content map	slide_map_id, slide_file_id, mapped_content_id
qc_log	validation findings	qc_id, entity_type, entity_id, result_status
changelog	immutable change history	change_id, entity_id, change_type
chunk_manifest	chunk continuation ledger	chunk_id, chunk_order, id_range_start, id_range_end

Detailed schema

The tables below are the **schema specification**. Types are restricted to `string`, `enum`, `integer`, `date_iso`, `datetime_iso`, `boolean`, and `long_text`.

Curriculum and foundations sheets

Sheet	Column definitions
readme	`workbook_version:string; schema_version:string; build_date:date_iso; build_notes:long_text; source_priority_note:long_text; assumption_note:long_text`
data_dictionary	`dict_id:string; sheet_name:string; column_name:string; data_type:enum; allowed_values:long_text; description:long_text; required:boolean; primary_key:boolean; foreign_key_to:string; example_value:string`

Sheet	Column definitions
taxonomy	`taxonomy_id:string; parent_taxonomy_id:string; taxonomy_type:enum(domain|subdomain|topic|age_group|setting|phase|special_population|operation|asset_type); level_scope:enum(shared|emt|aemt|paramedic|mixed); code:string; display_name:string; description:long_text; sort_order:integer; active:boolean`
modules	`module_id:string; taxonomy_id:string; level:enum(shared|emt|aemt|paramedic); module_type:enum(foundations|clinical|operations|review|assessment); title:string; short_summary:long_text; recommended_order:integer; est_minutes:integer; national_vs_local_flag:enum(core|variation_possible|local_override_needed)`
lessons	`lesson_id:string; module_id:string; taxonomy_id:string; level:enum(shared|emt|aemt|paramedic); title:string; short_summary:long_text; why_it_matters:long_text; must_know_points:long_text; red_flags:long_text; contraindications_or_dangers:long_text; common_mistakes:long_text; field_pearls:long_text; difficulty_1_to_5:integer; verification_status:enum(draft|reviewed|approved|pending_manual_review|deprecated); review_date:date_iso; national_vs_local_flag:enum(core|variation_possible|local_override_needed); deprecated_flag:boolean`
lesson_objectives	`objective_id:string; lesson_id:string; bloom_level:enum(remember|understand|apply|analyse|evaluate|create); objective_text:long_text; priority_rank:integer; assessed_by:string`
foundations	`foundation_id:string; cluster:enum(ap|terminology|abbreviations|pathophysiology|pharmacology|med_math|communication_documentation|legal_ethics|public_health|lifespan|ebm|safety_wellness); title:string; short_summary:long_text; core_concepts:long_text; mnemonic:string; difficulty_1_to_5:integer; revisit_priority:integer`
foundations_review_map	`frm_id:string; source_content_type:enum(lesson|skill|drug|ecg|calculation|scenario|question|media); source_content_id:string; foundation_id:string; relation_type:enum(prerequisite|revisit_if_missed|background|terminology_support|math_support); why_this_foundation_matters_here:long_text; trigger_threshold:integer; weak_area_followup:long_text`

Sheet	Column definitions
remediation_rules	`rule_id:string; trigger_type:enum(question_missed|scenario_failed|score_below_threshold|manual_assign|time_decay); level_scope:enum(shared|emt|aemt|paramedic|mixed); condition_expression:long_text; action_type:enum(assign_foundation|assign_lesson|assign_drill|assign_scenario_cluster); target_content_type:string; target_content_id:string; cooldown_days:integer; active:boolean`
glossary	`glossary_id:string; term:string; expansion:string; definition:long_text; preferred_term_flag:boolean; level_scope:enum(shared|emt|aemt|paramedic|mixed); notes:long_text`

Skills, drugs, ECG, maths, scenarios, and questions

Sheet	Column definitions
skills	`skill_id:string; taxonomy_id:string; title:string; level_min:enum(emt|aemt|paramedic); indications:long_text; contraindications:long_text; equipment_needed:long_text; preparation_steps:long_text; critical_fail_points:long_text; safety_points:long_text; reassessment_points:long_text; documentation_points:long_text; psychomotor_complexity:integer; verification_status:enum(draft|reviewed|approved|pending_manual_review)`
skill_steps	`skill_step_id:string; skill_id:string; step_order:integer; action_text:long_text; must_complete:boolean; branch_if_failed:string; media_hint:string`
drug_classes	`drug_class_id:string; class_name:string; description:long_text; common_examples:long_text; notes:long_text`
drugs	`drug_id:string; drug_class_id:string; generic_name:string; brand_names:string; class_name:string; level_min:enum(emt|aemt|paramedic); indications:long_text; contraindications:long_text; adult_dose_text:long_text; paediatric_dose_text:long_text; route_text:string; repeat_dose_rules:long_text; onset_text:string; duration_text:string; adverse_effects:long_text; monitoring:long_text; protocol_variation_note:long_text; verification_status:enum(draft|reviewed|approved|pending_manual_review)`

Sheet	Column definitions
ecg_bank	`ecg_id:string; taxonomy_id:string; rhythm_name:string; ecg_type:enum(rhythm_strip|twelve_lead|waveform_case); difficulty_1_to_5:integer; source_leads:string; strip_duration_sec:integer; identification_pearls:long_text; common_confusions:long_text; immediate_priorities:long_text; treatment_considerations_by_level:long_text; verification_status:enum(draft|reviewed|approved|pending_manual_review)`
calculation_drills	`calc_id:string; taxonomy_id:string; calc_type:enum(unit_conversion|mg_per_kg|ml_dose|drip_rate|pump_rate|weight_conversion|concentration|paediatric_safety); level_min:enum(emt|aemt|paramedic); stem:long_text; formula_text:long_text; worked_solution:long_text; common_traps:long_text; difficulty_1_to_5:integer; verification_status:enum(draft|reviewed|approved)`
scenarios	`scenario_id:string; taxonomy_id:string; level:enum(emt|aemt|paramedic); age_group:enum(neonate|infant|toddler|preschool|school_age|adolescent|adult|older_adult); special_population_flags:long_text; setting:enum(home|public_place|roadway|industrial|school|long_term_care|clinic|hospital|wilderness|water|aircraft|interfacility|mass_gathering); dispatch_information:long_text; scene_cues:long_text; safety_hazards:long_text; initial_impression:long_text; difficulty_1_to_5:integer; terminal_outcome_type:enum(stabilise|deteriorate|arrest|handover|refusal|debrief_only); verification_status:enum(draft|reviewed|approved)`
scenario_steps	`step_id:string; scenario_id:string; phase:enum(dispatch|en_route|arrival|scene_size_up|primary|immediate_interventions|secondary|differential|ongoing_management|reassessment|transport|handoff|debrief); cue_text:long_text; data_payload_json:long_text; expected_action:long_text; scope_allowed:long_text; dangerous_wrong_action:long_text; consequence_text:long_text; next_default_step_id:string`
scenario_branches	`branch_id:string; scenario_id:string; from_step_id:string; trigger_action_pattern:long_text; branch_condition:long_text; to_step_id:string; consequence_severity:enum(low|moderate|high|critical); debrief_note:long_text`

Sheet	Column definitions
questions	`question_id:string; taxonomy_id:string; level:enum(emt|aemt|paramedic); scenario_id_nullable:string; ecg_id_nullable:string; item_type:enum(single_best_answer|multiple_response|build_list|drag_drop|hotspot|graphical_ecg|waveform_interpretation|scenario_cluster|clinical_judgment); difficulty_1_to_5:integer; cognitive_level:enum(recall|application|analysis|judgment); stem:long_text; answer_format:string; correct_answer_key:string; common_trap:long_text; remediation_rule_id_nullable:string; national_vs_local_flag:enum(core|variation_possible|local_override_needed); verification_status:enum(draft|reviewed|approved|pilot_ready|deprecated)`
question_options	`option_id:string; question_id:string; option_key:string; option_text:long_text; display_order:integer; is_correct:boolean`
question_rationales	`rationale_id:string; question_id:string; option_id_nullable:string; rationale_type:enum(correct|incorrect_option|teaching_point); rationale_text:long_text`
psychometrics	`psy_id:string; question_id:string; item_status:enum(draft|reviewed|pilot|live|retired); pilot_status:enum(not_started|ready|piloting|complete); target_blueprint_domain:string; scope_review_status:enum(pending|passed|flagged); bias_review_status:enum(pending|passed|flagged); readability_status:enum(pending|passed|flagged); p_value_target:string; discrimination_index:string; last_item_review_date:date_iso`

References, joins, media, overrides, and governance

Sheet	Column definitions
references	`reference_id:string; source_type:enum(scope|education_standard|instructional_guidance|test_plan|handbook|guideline|dataset|toolkit|local_protocol|other); organisation:string; title:string; edition_or_year:string; section_or_page_if_known:string; jurisdiction:string; authority_level:integer; last_verified:date_iso; licence_note:string`
content_links	`link_id:string; from_type:string; from_id:string; to_type:string; to_id:string; relation_type:string; sort_order:integer`
content_reference_links	`cref_id:string; content_type:string; content_id:string; reference_id:string; pinpoint_text:string; citation_priority:integer`

Sheet	Column definitions
media_manifest	`media_id:string; asset_type:enum(ecg_raw|ecg_image|heart_sound|lung_sound|monitor_waveform|image|video|document|link_only); source_name:string; source_asset_id:string; file_name:string; folder_path:string; mime_type:string; sample_rate_hz:integer; channels:integer; duration_ms:integer; image_width_px:integer; image_height_px:integer; waveform_type:string; body_site:string; clinical_label:string; ecg_id_nullable:string; checksum_sha256:string; anonymised_flag:boolean; licence_name:string; licence_url:string; attribution_text:long_text; redistribution_allowed:boolean; derivative_allowed:boolean; commercial_use_allowed:boolean; alt_text:long_text; caption:long_text; verification_status:enum(draft|reviewed|approved|permission_required|link_only|rejected); review_date:date_iso`
media_links	`mlink_id:string; media_id:string; content_type:string; content_id:string; usage_role:enum(primary_teaching_image|question_stem|question_option|scenario_monitor|reference_only|flashcard); display_order:integer`
protocol_overrides	`override_id:string; jurisdiction_type:enum(state|territory|agency|service|medical_director); jurisdiction_name:string; content_type:string; content_id:string; override_type:enum(scope|dose|indication|contraindication|destination|equipment|documentation|wording); override_text:long_text; effective_date:date_iso; expiry_date:date_iso; approval_reference_id:string; active:boolean`
slide_inventory	`slide_file_id:string; source_path:string; file_name:string; file_type:string; last_modified:datetime_iso; slide_count:integer; checksum_sha256:string; rights_status:enum(user_provided|unknown|permission_required); ingest_status:enum(queued|parsed|mapped|needs_ocr|error)`
slide_mapping	`slide_map_id:string; slide_file_id:string; slide_number:integer; extracted_title:string; extracted_topics:long_text; mapped_taxonomy_id:string; mapped_content_type:string; mapped_content_id:string; confidence_score:integer; human_review_required:boolean`

Sheet	Column definitions
qc_log	`qc_id:string; entity_type:string; entity_id:string; check_type:string; result_status:enum(pass|warn|fail); finding_text:long_text; reviewer_role:string; reviewed_at:datetime_iso; remediation_ticket:string`
changelog	`change_id:string; entity_type:string; entity_id:string; change_type:enum(create|update|fix|deprecate|supersede); change_summary:long_text; changed_by_role:string; changed_at:datetime_iso; supersedes_id_nullable:string`
chunk_manifest	`chunk_id:string; chunk_order:integer; chunk_name:string; entity_scope:string; id_range_start:string; id_range_end:string; export_file_name:string; checksum_sha256:string; released_at:datetime_iso`

Relationship diagram

```
erDiagram
    TAXONOMY ||--o{ MODULES : organises
    MODULES ||--o{ LESSONS : contains
    LESSONS ||--o{ LESSON_OBJECTIVES : has
    FOUNDATIONS ||--o{ FOUNDATIONS_REVIEW_MAP : supports
    LESSONS ||--o{ CONTENT_LINKS : links
    SKILLS ||--o{ CONTENT_LINKS : links
    DRUGS ||--o{ CONTENT_LINKS : links
    ECG_BANK ||--o{ CONTENT_LINKS : links
    CALCULATION_DRILLS ||--o{ CONTENT_LINKS : links
    SCENARIOS ||--o{ SCENARIO_STEPS : contains
    SCENARIOS ||--o{ SCENARIO_BRANCHES : branches
    QUESTIONS ||--o{ QUESTION_OPTIONS : has
    QUESTIONS ||--o{ QUESTION_RATIONALES : explains
    QUESTIONS ||--o{ PSYCHOMETRICS : measures
    REFERENCES ||--o{ CONTENT_REFERENCE_LINKS : supports
    MEDIA_MANIFEST ||--o{ MEDIA_LINKS : attaches
    SLIDE_INVENTORY ||--o{ SLIDE_MAPPING : maps
    PROTOCOL_OVERRIDES }o--|| REFERENCES : approved_by
```

Coverage and assessment design

Current public NREMT blueprints make two design choices impossible to ignore. First, the examinations are not weighted evenly; clinical judgement dominates AEMT and Paramedic, while EMT is weighted heavily toward scene safety, primary assessment, treatment/transport, and operations. Second, paediatric care is integrated across the content rather than isolated in one paediatric domain. Public EMSC readiness resources also point out that many EMS agencies see relatively few paediatric patients and that deliberate paediatric readiness is needed, including equipment, coordinators, skills checking, and disaster planning. That is why this design intentionally **over-represents paediatric scenarios and remediation loops** compared with raw call frequency. [13]

Scenario taxonomy and coverage map

Use five taxonomy dimensions plus one overlay system:

- **Age groups**: `neonate`, `infant`, `toddler`, `preschool`, `school_age`, `adolescent`, `adult`, `older_adult`
- **Settings**: `home`, `public_place`, `roadway`, `industrial`, `school`, `long_term_care`, `clinic`, `hospital`, `wilderness`, `water`, `aircraft`, `interfacility`, `mass_gathering`
- **Phases**: `dispatch`, `en_route`, `arrival`, `scene_size_up`, `primary`, `immediate_interventions`, `secondary`, `differential`, `ongoing_management`, `reassessment`, `transport`, `handoff`, `debrief`
- **Special-population overlays**: paediatric, geriatric, obstetric, neonatal, bariatric, developmental disability/autism, behavioural crisis, hospice/palliative, technology dependence, language barrier, immunocompromised, intoxication, custody/law enforcement interface
- **Operations overlays**: MCI/START or local triage, hazmat awareness, traffic incident management, rescue awareness, violent scene awareness, disaster shelter/public-health interface, documentation/radio/handoff
- **Coverage rule**: every major domain must include all applicable settings/phases across all three levels, with more paediatric repetition than adult per topic cluster

Initial content targets by domain and level

The target split below is a **build recommendation** informed by the public blueprint distributions and the need to deliberately strengthen paediatric and ECG-heavy content. [14]

Domain	EMT lessons / scenarios / questions	AEMT lessons / scenarios / questions	Paramedic lessons / scenarios / questions
Foundations and preparatory	36 / 30 / 180	18 / 15 / 90	24 / 20 / 120
Airway, respiration, ventilation	24 / 45 / 160	16 / 30 / 110	26 / 45 / 180
Cardiology, resuscitation, ECG	22 / 35 / 150	18 / 30 / 120	34 / 55 / 260
Trauma	28 / 55 / 190	16 / 30 / 100	28 / 50 / 200
Medical, OB/GYN, special populations	38 / 80 / 260	24 / 50 / 180	44 / 85 / 320
Operations, MCI, safety, documentation	22 / 35 / 100	13 / 15 / 40	15 / 20 / 80
Totals	**170 / 280 / 1,040**	**105 / 170 / 640**	**171 / 275 / 1,160**

That v1 recommendation yields roughly **446 lessons**, **725 scenarios**, and **2,840 original questions**. If you want a leaner first release, trim evenly to about **300 lessons**, **500 scenarios**, **2,000 questions**, and **200 media assets** while preserving the same proportions.

Question-bank design

NREMT public examination materials say that all levels now include technology-enhanced items, and that AEMT/Paramedic examinations include scenario-linked clinical-judgement items. AEMT is weighted 31–35% to Clinical Judgment and Paramedic 34–38%; public NREMT material also states that 30% of the Paramedic Cardiology and Resuscitation domain is graphical ECG content. That means the workbook should not be built as a multiple-choice bank with a few ECG pictures bolted on later. It needs native support for interactive and graphical item classes from day one. [15]

Required item types:

`single_best_answer`, `multiple_response`, `build_list`, `drag_drop`, `hotspot`, `graphical_ecg`, `waveform_interpretation`, `scenario_cluster`, `clinical_judgment`

Difficulty scale:

Difficulty	Definition
1	Foundational recall or simple identification
2	Straightforward application with minimal ambiguity
3	Moderate application requiring prioritisation
4	Complex case reasoning with competing cues
5	High-uncertainty clinical judgement under time pressure

Required item fields:

`question_id`, `taxonomy_id`, `level`, `scenario_id_nullable`, `ecg_id_nullable`, `item_type`, `difficulty_1_to_5`, `cognitive_level`, `stem`, `answer_format`, `correct_answer_key`, `why_correct`, `why_each_wrong`, `reference_ids`, `remediation_rule_id`, `national_vs_local_flag`, `verification_status`

Rationale rules:

- every item must have **one correct-answer rationale**
- every distractor must have **its own wrong-answer rationale**
- scenario-cluster items must also have a **cluster teaching note**
- all rationales must reference the same official source layer used by the lesson
- if an option is locally variable, the rationale must explicitly say so and set `national_vs_local_flag != core`

Psychometric/QC fields:

`item_status`, `pilot_status`, `target_blueprint_domain`, `scope_review_status`, `bias_review_status`, `readability_status`, `p_value_target`, `discrimination_index`, `last_item_review_date`

NREMT public materials say exam items are written by EMS subject-matter experts, include unscored pilot items, and that poorly performing questions are removed, reviewed, revised, and repiloted. Copy that process in miniature for your own bank. [16]

Media, governance, and export

For waveform and audio assets, prioritise primary public repositories over prettified screenshots. MIT-BIH offers open-access rhythm data with reference annotations and ODC-By licensing; PTB-XL offers a large open 12-lead ECG dataset with unique `ecg_id` values, pseudonymised metadata, 500 Hz waveform files plus 100 Hz downsampled files, and CC BY 4.0 licensing; CirCor DigiScope is a large paediatric heart-sound dataset on PhysioNet with open licensing; VitalDB provides de-identified high-resolution perioperative waveforms and clinical data under a data-use agreement. AHA guidance is essential for content but its own site points users to copyright-permission procedures, so AHA figures/algorithms should not be bulk-ingested as reusable media without permission. [17]

Media asset folder structure and metadata fields

Folder	Purpose	Required metadata fields
`/raw/ecg/physionet/mitdb/`	raw rhythm-strip source files	media_id, source_name, source_asset_id, clinical_label, sample_rate_hz, licence_name
`/raw/ecg/physionet/ptbxl/`	raw 12-lead WFDB ECG files	media_id, ecg_id, source_asset_id, lead_set, sample_rate_hz, duration_ms
`/raw/audio/heart/physionet/circor/`	paediatric heart-sound WAV files	media_id, body_site, clinical_label, sample_rate_hz, duration_ms
`/raw/audio/lung/unspecified_pending/`	lung sounds pending licence review	media_id, source_name, permission_status, review_date
`/raw/waveforms/vitaldb/`	raw monitor waveform files	media_id, waveform_type, case_id_pseudonym, sample_rate_hz, anonymised_flag
`/derived/ecg/png/`	rendered ECG strips/12-leads	media_id, parent_media_id, image_width_px, image_height_px, caption
`/derived/audio/mp3/`	app-streamable audio derivatives	media_id, parent_media_id, duration_ms, caption, alt_text
`/derived/waveforms/png/`	rendered capnogram/pleth/ABP images	media_id, parent_media_id, waveform_type, image_width_px, image_height_px
`/manifests/`	CSV/JSON manifests and checksums	manifest_version, schema_version, created_at, row_count

Import-ready `media_manifest` schema

Use this exact minimum schema:

`media_id`, `asset_type`, `source_name`, `source_asset_id`, `file_name`, `folder_path`, `mime_type`, `sample_rate_hz`, `channels`, `duration_ms`, `image_width_px`, `image_height_px`, `waveform_type`, `body_site`, `clinical_label`, `ecg_id_nullable`, `checksum_sha256`, `anonymised_flag`, `licence_name`, `licence_url`, `attribution_text`, `redistribution_allowed`, `derivative_allowed`, `commercial_use_allowed`, `alt_text`, `caption`, `verification_status`, `review_date`

Manifest example rows

media_id	asset_type	source_name	source_asset_id	file_na
media_ecg_000001	ecg_raw	PhysioNet MIT-BIH	rec_100	100.da
media_ecg_000142	ecg_raw	PhysioNet PTB-XL	00001_hr	00001
media_audio_000021	heart_sound	PhysioNet CirCor	50782_MV_1	50782
media_wf_000011	monitor_waveform	VitalDB	case_012984	case_0
media_audio_000090	lung_sound	unspecified_primary_source_pending_review	pending	pendi

ECG and audio asset requirements

Use **raw masters + derived app files**. For ECG, preserve source waveform formats where possible; render derivatives as PNG or WebP at roughly **1600–2400 px width** so users can zoom without smearing thin lines. For audio, preserve raw WAV/PCM when available; produce MP3 or Ogg derivatives only for streaming. Always record the original sampling rate in the manifest instead of normalising everything blindly. Known primary-source examples: MIT-BIH uses 360 samples/second per channel; PTB-XL provides 500 Hz waveform files and also 100 Hz downsampled versions; VitalDB describes 500 Hz waveform data and numeric values at 1–7 second intervals. [18]

Require these checks before an asset is ship-ready:

- `anonymised_flag = true` or a source document proving de-identification/pseudonymisation
- `licence_name` populated and human-reviewed
- `redistribution_allowed` confirmed
- `alt_text` and `caption` written in plain English
- `linked ecg_id/media_id` present for ECG-linked teaching items

- `checksum_sha256` present for reproducibility
- `verification_status != approved` if any rights or anonymisation uncertainty remains

PTB-XL explicitly documents pseudonymised identifiers and HIPAA-style handling for very old ages and shifted dates; VitalDB states that the open dataset uses random case IDs and de-identified subject IDs. [19]

Filename examples:

- `ecg_mitdb_rec100_rhythmstrip_sinus_360hz_odcby_v1.png`
- `ecg_ptbxl_ecg_00001_12lead_scp-rhythm_500hz_ccby4_v1.png`
- `audio_circor_50782_mv1_systolic-murmur_mitral_4000hz_odcby_v1.wav`
- `wf_vitaldb_case012984_capnogram_low-etco2_500hz_dua_v1.png`

Metadata examples:

- `clinical_label = narrow_complex_tachycardia`
- `body_site = mitral_area`
- `waveform_type = capnogram`
- `alt_text = six-second lead II rhythm strip showing regular narrow-complex tachycardia`
- `caption = Mitral-area systolic murmur audio, paediatric dataset, use for auscultation pattern recognition`

Governance and QC rules

The governance layer should be blunt and boring, which is exactly what you want in medical content. Every clinically meaningful row should carry: `verification_status`, `review_date`, `reviewer_role`, `source_conflict_flag`, `national_vs_local_flag`, and `deprecated_flag`. Every override should carry an approval trace. Every change should be immutable in `changelog`. Public NREMT material describes SME writing, pilot items, psychometric review, and recycling/removal of poor items; EMS.gov and NASEMSO emphasise evidence-based guideline development and performance management. Your workbook governance should mirror that discipline. [20]

Required governance fields:

- `verification_status`: `draft | reviewed | approved | pending_manual_review | deprecated`
- `review_date`
- `reviewer_role`: `educator | medical_director | editor | data_engineer | psychometric_reviewer`
- `source_conflict_flag:boolean`
- `national_vs_local_flag`: `core | variation_possible | local_override_needed`
- `approval_reference_id`
- `deprecated_flag:boolean`

Approval workflow:

`draft -> automated validation -> educator review -> clinical scope review -> protocol/local review if needed -> editorial clean-up -> approved -> exported`

Conflict handling:

- same topic, different source tiers: higher-priority tier wins
- same source tier, newer source supersedes older
- unresolved conflict: keep both in notes, set `source_conflict_flag = true`, and withhold `approved`
- protocol-specific conflict: move the difference into `protocol_overrides`, not the national-core rows

Deterministic IDs, chunking, and export

Use deterministic prefixes and fixed-width counters:

- `tax_000001`
- `mod_emt_000001`
- `les_prm_000001`
- `fdn_000001`
- `scn_aemt_000001`
- `q_emt_000001`
- `media_ecg_000001`
- `ref_000001`
- `ovr_ca_000001`

Chunking rules:

- preserve schema and IDs across all chunks
- append only
- any correction goes into `changelog`
- every chunk gets a `chunk_manifest` row with ID ranges and checksums
- chunk names should be stable, for example `chunk_core_taxonomy`, `chunk_emt`, `chunk_aemt`, `chunk_paramedic`, `chunk_media`, `chunk_protocol_overrides`

Exports:

- **XLSX** for human review
- **Google Sheets** compatible workbook
- **per-sheet CSV** for ETL/import
- **JSONL** for app loaders, embeddings, or retrieval indexes
- **media manifest** as CSV + JSON

Packaging and timeline

Builder output should include: **the multi-sheet workbook**, **a PDF that contains the polished master prompt plus this implementation report for the Replit Agent**, and **media-manifest CSV/JSON exports**. If the dataset is too large, the builder must ship sequential chunks while preserving all existing IDs. That packaging choice is consistent with Replit Agent's project-generation workflow and with OpenAI's server-side API patterns for file-backed or retrieval-backed application logic. [21]

```
flowchart LR
    A[Acquire official sources and user slides] --> B[Build references and
taxonomy]
```

```
    B --> C[Create modules, lessons, foundations]
    C --> D[Create scenarios and branches]
    C --> E[Create questions and rationales]
    A --> F[Ingest media and licences]
    D --> G[Link content via joins]
    E --> G
    F --> G
    G --> H[Automated validation and QC log]
    H --> I[Educator and medical-director review]
    I --> J[Protocol override layer]
    J --> K[XLSX and Google Sheets]
    J --> L[Per-sheet CSV]
    J --> M[JSONL and media manifests]
    I --> N[Changelog and chunk manifest]
```

Source priorities, slide ingestion, and Replit integration

Prioritised source list and conflict handling

Use this source stack in order:

1. National EMS Scope of Practice Model
2. National EMS Education Standards
3. Official EMT/AEMT/Paramedic educational guidance and companion instructional material
4. Current NREMT public test plans, handbooks, and sample packets
5. Current AHA CPR/ECC and first aid guidance
6. NASEMSO National Model EMS Clinical Guidelines and EMS.gov evidence-based guideline resources
7. HRSA/EMSC paediatric-readiness resources
8. User-provided Google Drive slides
9. Other primary sources when needed

Why this order: the scope model explicitly leaves actual scope to states and territories while providing the national foundation; the Education Standards define minimum competency rather than a single curriculum; NREMT public plans define what the exams emphasise; AHA guidance drives resuscitation specifics; NASEMSO/EMS.gov material supports protocol logic and evidence-based care; paediatric-readiness resources fill a known training gap. [22]

Actionable checklist for the Google Drive slides

The slide folder should be treated as a **user-provided teaching source**, not a trump card over newer official material.

1. Create one `slide_inventory` row per file with `slide_file_id`, `source_path`, `file_name`, `file_type`, `last_modified`, `slide_count`, `checksum_sha256`, and `ingest_status = queued`.
2. Extract visible text, slide titles, notes, diagrams, tables, and chapter cues per slide. If OCR is required, set `ingest_status = needs_ocr` and log OCR confidence.
3. Generate thumbnails/contact sheets for quick educator review.

4. Infer likely `taxonomy_id` and `module_id` from slide titles/chapters; write draft mappings into `slide_mapping`.
5. If one slide maps to multiple topics, create multiple `slide_mapping` rows rather than stuffing lists into one cell.
6. Mark each slide-derived mapping as either `supporting_teaching_source`, `unique_example_asset`, or `rights_unclear`.
7. If a slide conflicts with official national/current guidance, create a `qc_log` entry and preserve the official source as canonical.
8. For monitor screenshots or diagrams in the slides, check whether the rights allow redistribution; if unknown, generate `media_manifest` rows with `verification_status = permission_required`.
9. Have an educator review the mapped slides for chapter order, duplicates, and obsolete content.
10. After approval, create `content_links` from slides to lessons, scenarios, questions, and media for app-side "study from your textbook slides" navigation.

This step is important because official EMS education documents are standards, not a turnkey curriculum, and your user-provided slide deck can add pacing, chapter structure, and local teaching flavour once it has been reconciled against current official material. [23]

Replit Agent and API integration notes

For deployment via Replit [24], the cleanest path is: use Replit Agent to scaffold the app and admin tools; store the workbook as versioned CSV/JSON plus a database mirror; and implement "AI practice mode" as **retrieval grounded on the workbook content** rather than trying to train a model from scratch. If you later want fine-tuning, do it only on your own original content, never on copyrighted exam items. Replit Agent is designed to create and deploy apps from natural-language instructions, and Replit Secrets stores encrypted environment variables for API keys. Agent-generated apps are not compatible with static deployments, so start with Autoscale for variable traffic or Reserved VM for steadier production needs; exact costs are live-priced and should be treated as unspecified until checked on Replit's current pricing pages. [25]

For the OpenAI [26] side, the app should use server-side API calls only. OpenAI's docs say API keys are secret, must not be exposed in client-side code, and should be loaded from environment variables or a key-management service. So the Replit Agent should explicitly prompt the user to enter their API key, then store it in Replit Secrets, never in the workbook, frontend bundle, or repository. If the AI mode uses workbook-backed retrieval, index the per-sheet CSV/JSON exports into your chosen retrieval layer; if using OpenAI file-backed retrieval patterns, keep the workbook content limited to original materials you own or can redistribute. [27]

A low-cost implementation pattern is:

- workbook as the source of truth
- exported CSV/JSON mirrored into SQLite or PostgreSQL
- admin UI in Replit for content filtering/QC
- practice UI uses workbook-grounded retrieval for explanations and adaptive drills
- optional local/lightweight inference for non-clinical tasks like tagging, duplicate detection, and metadata enrichment
- server-side OpenAI calls only for the practice/explanation layer
- no copyrighted exam items anywhere in prompts, storage, or fine-tuning sets

Open questions and limitations

The provided Google Drive folder was referenced and incorporated into the builder instructions, but the slide contents were not directly accessible from this environment, so the ingestion section is procedural rather than executed.

For lung-sound assets, an equivalent primary/open repository as clear-cut as PhysioNet's ECG and CirCor resources was not established in this pass. For that class, use `source_name = unspecified_primary_source_pending_review` until a human confirms licensing and redistribution rights.

Primary guidance sources are strong for scope, education, public exam alignment, resuscitation, evidence-based protocols, paediatric readiness, ECG, heart sounds, and monitor waveforms. The main remaining manual-review risk is **local protocol variation**: especially medication dosing, advanced airway limits, destination rules, and procedure authorisation. Those belong in `protocol_overrides`, not in the national-core rows. [28]

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[1] [11] [22] [26] [28] https://www.ems.gov/assets/National_EMS_Scope_of_Practice_Model_2019.pdf
https://www.ems.gov/assets/National_EMS_Scope_of_Practice_Model_2019.pdf

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[3] [12] [23] https://www.ems.gov/resources/newsletters/january-2021/what-the-new-ems-education-standards-will-mean-for-ems/
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